

OVERVIEW .ai

OV20i AI Vision Inspection

Stadler St. Margrethen – Connector Inspection Demo

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OV20i Benefits and Advantages



100% Coverage

Every part inspected with consistent accuracy, eliminating human error



Sub-Second Inspection

Fast cycle times with real-time AI inference on the edge



5 Min Model Training

Go from zero to production-ready AI model in under an hour



Deep Learning AI

Vision transformer technology trains with as few as 5 images



No Software License

No recurring fees, no external PC needed. Everything runs on-device



Built-in Lighting

Configurable LED lighting with photometric control, no external light needed

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A classification recipe built in 45 minutes from a single image

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A Node-RED dashboard built in 1 hour, plus easy integration

01

Introduction

Stadler Connector Inspection Demo

One AI classification recipe was created and demonstrated during the visit to Stadler in St. Margrethen, Switzerland on June 10, 2026. The OV20i was tasked with verifying electrical connectors before they are plugged in the cabinet.

Classification Recipe – Female Connector

Primary Demo

- Female Pins – present, absent or obstructed
- Encoders (female/male) and Fixations (screw/thread)
- Encoder and fixation types are the only safeguard against plugging a connector into the wrong adjacent position
- Working recipe in 45 min – from a single image
- Custom operator dashboard built from scratch in 1 hour

02

Recipe Setup Process

From a single image to a working AI recipe in 45 minutes

Step 1: Imaging Setup

Exposure, motorized focus and the built-in LED lighting were tuned live on the connector. The guided workflow then walks through three key AI blocks:

Step 2: Template Image and Alignment

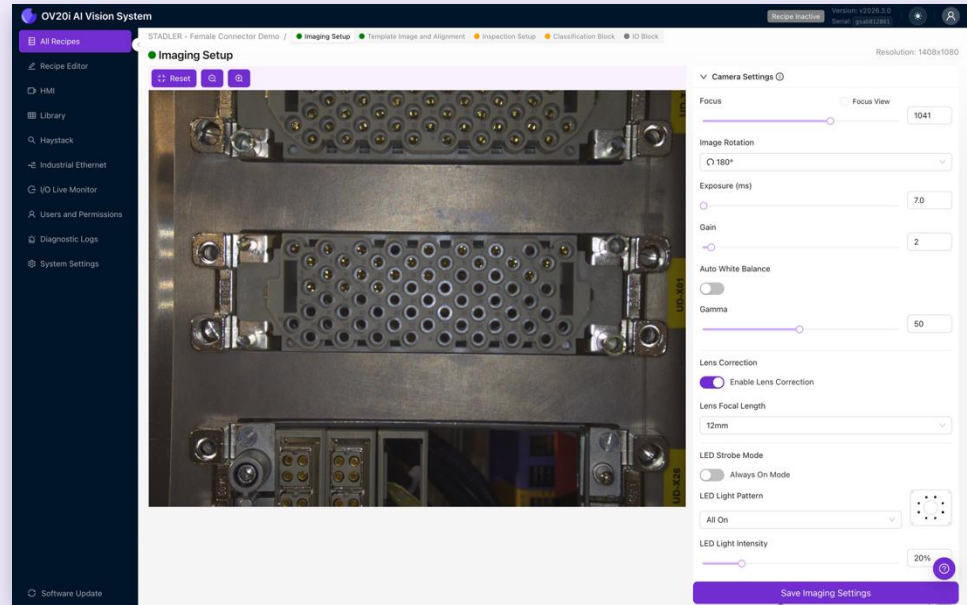
Track the part in the field of view

Step 3: Inspection Setup

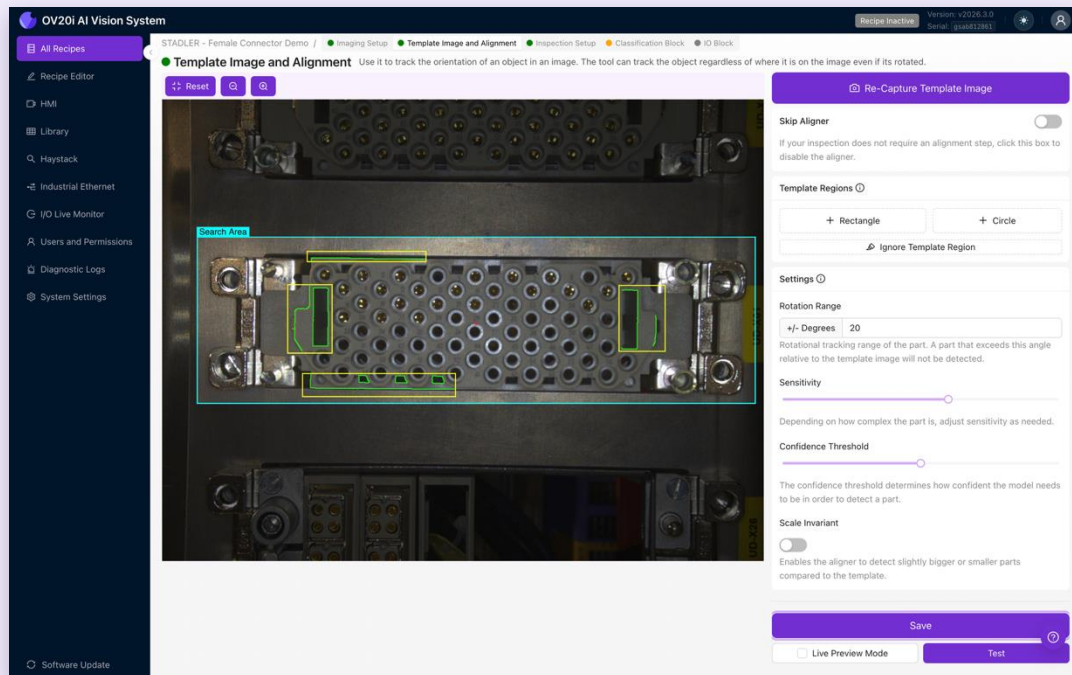
Define regions of interest (ROIs)

Steps 4–5: Label and Train

Annotate each ROI and train the AI model



Step 2: Template Image and Alignment



Part Tracking

The aligner uses recognizable features on the connector (housing corners and bracket landmarks) to track its position regardless of where it is in the image.

Settings:

- Template Regions (Rectangle/Circle)
- Rotation Range
- Sensitivity
- Confidence Threshold
- Scale Invariant option

Step 3: Inspection Setup – Define ROIs

One ROI was drawn for every female pin, encoder and fixation, grouped into three Inspection Types (Female Pins, Encoders, Fixations) — so every single position on the connector gets its own AI decision.

OV20i AI Vision System

STADLER - Female Connector Demo / ● Imaging Setup ● Template Image and Alignment ● **Inspection Setup** ● Classification Block ● IO Block

Inspection Setup Use it to define parts of the image that you want to focus on and make decisions.

Reset Search Bounding Boxes ROI Labels

Inspection Types

Inspection Type	# of ROIs	Edit
Female Pins	64	Edit
Encoders	4	Edit
Fixations	4	Edit

+ Add Inspection Type

Transformation Undo

Height: X: Width: Y: Angle:

Alignment Distribution

+ - | |

Multiple ROI Selection

Inspection Regions

+ Add Inspection Region

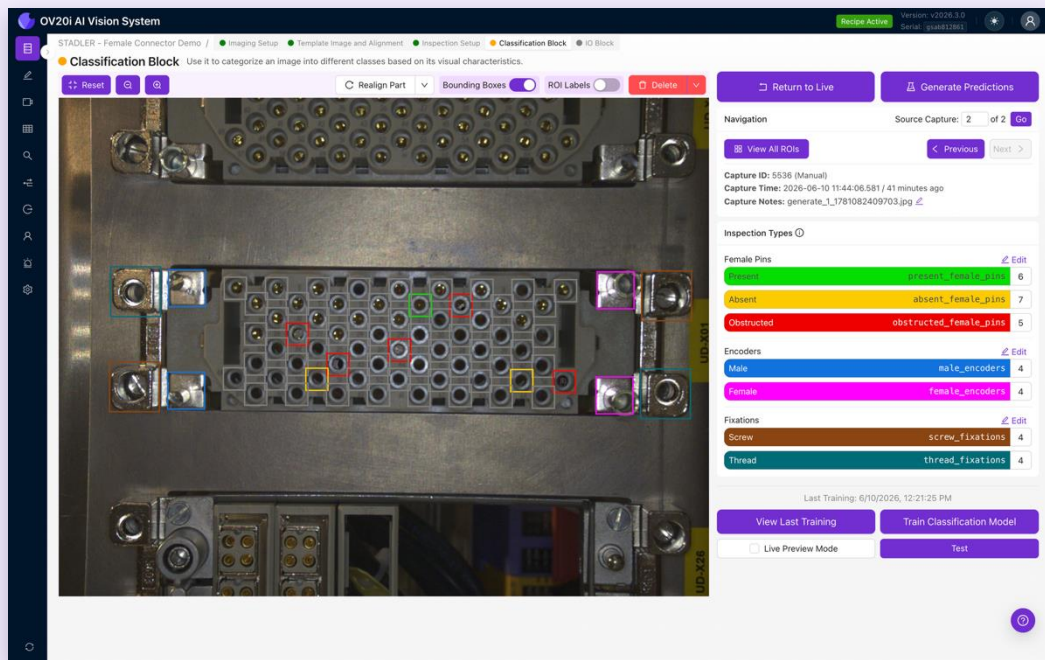
Ignore Regions

ROI Name	Inspection Type	Edit
2	Female Pins	Edit
3	Female Pins	Edit
4	Female Pins	Edit
5	Female Pins	Edit
6	Female Pins	Edit
7	Female Pins	Edit
8	Female Pins	Edit

Software Update

Step 4: Label with the Classification Block

Each ROI crop is labeled directly in the browser. One classification recipe handles all three checks at once, with dedicated classes per Inspection Type.

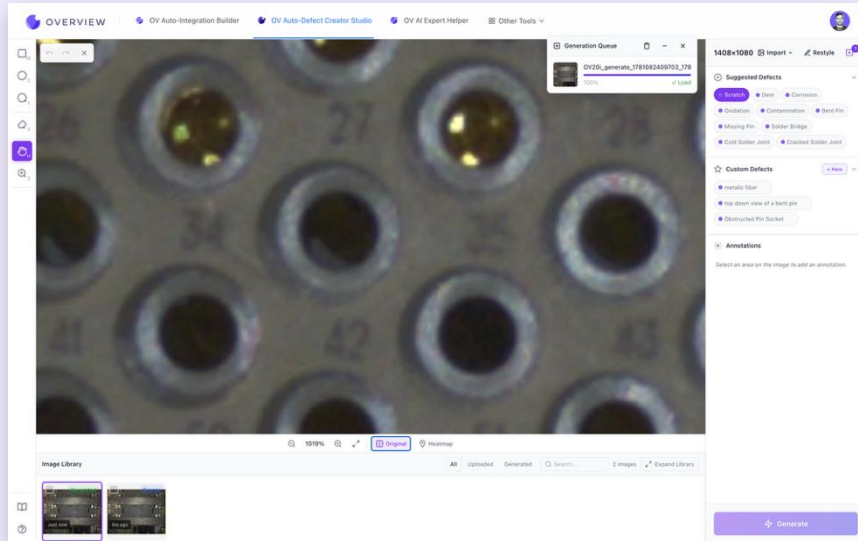


Classes: Female Pins (present / absent / obstructed) | Encoders (female / male) | Fixations (screw / thread)

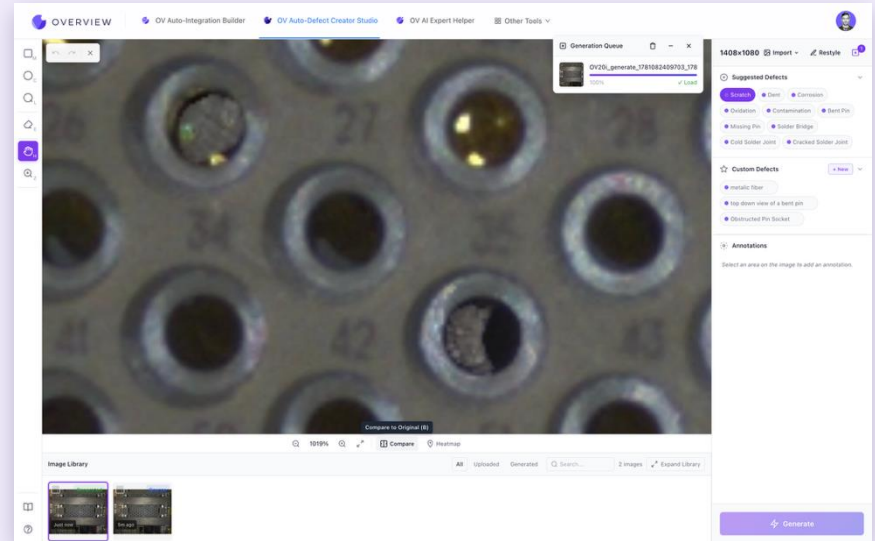
Generate Defects with AI Defect Creator

The OV Auto-Defect Creator Studio

Generates photorealistic synthetic defect images using AI. For the Stadler demo, obstructed female pins were created from the single captured image — no need to damage a real connector. Included free with the OV20i camera.



Original female pins

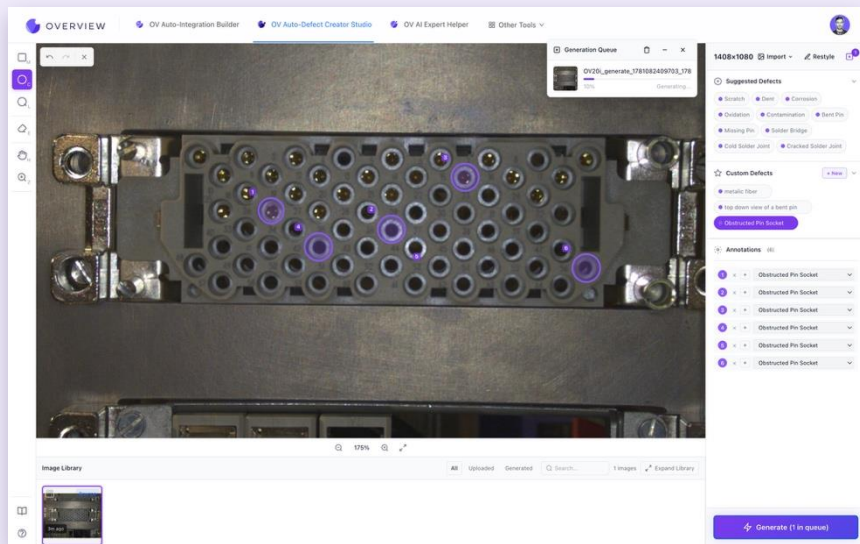


AI-generated obstructed pins

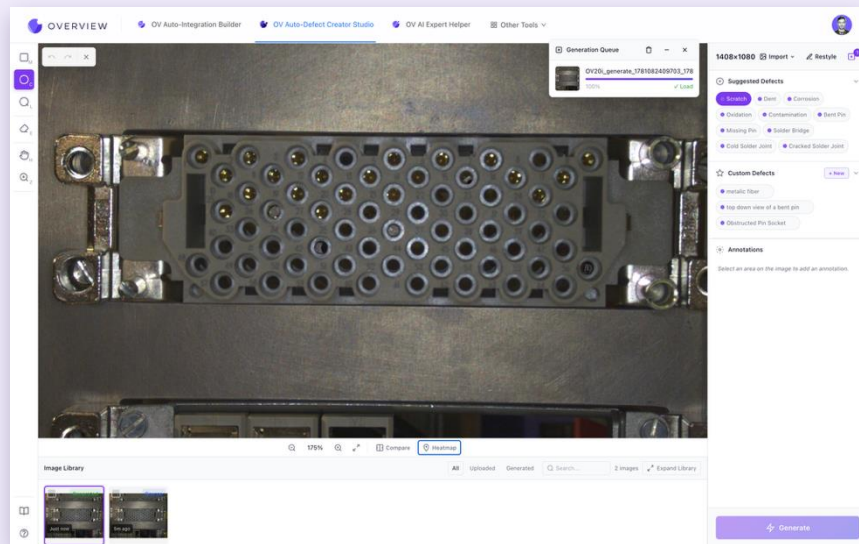
From One Good Image to a Full Training Set

Mark, generate, add to dataset

Defect locations are simply marked on the sockets to obstruct and the AI generates realistic variations. Within minutes the recipe had enough defective examples to train on — all created from one good part.



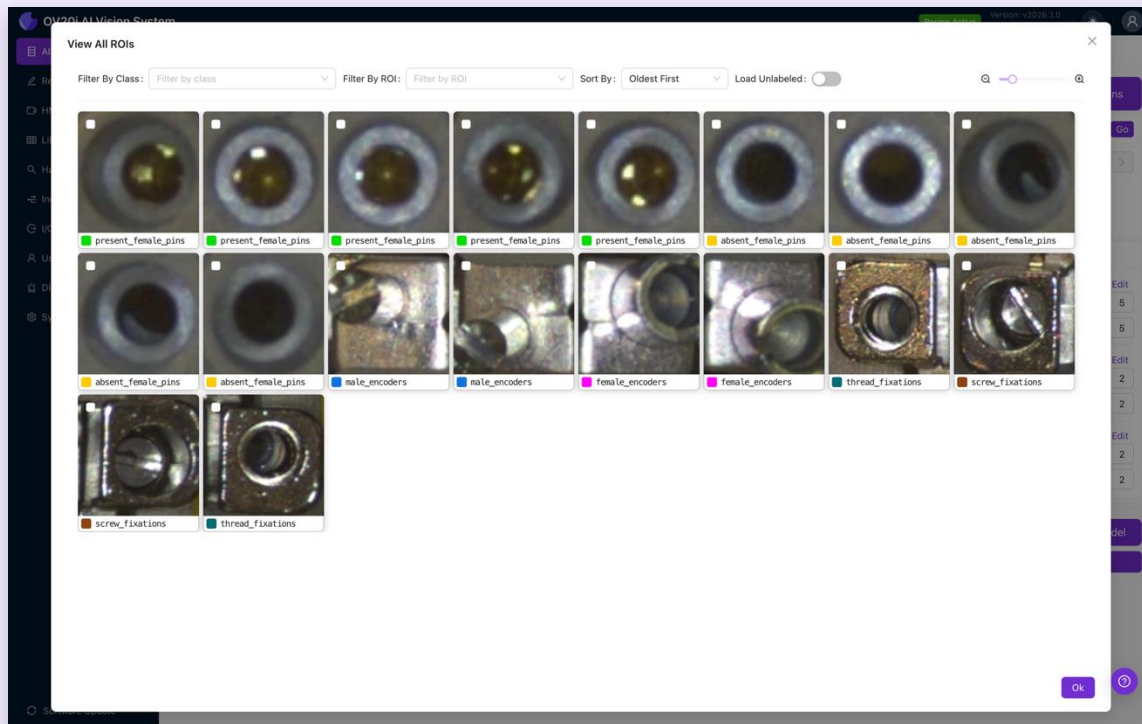
Defect locations marked for generation



Generated image added to the training dataset

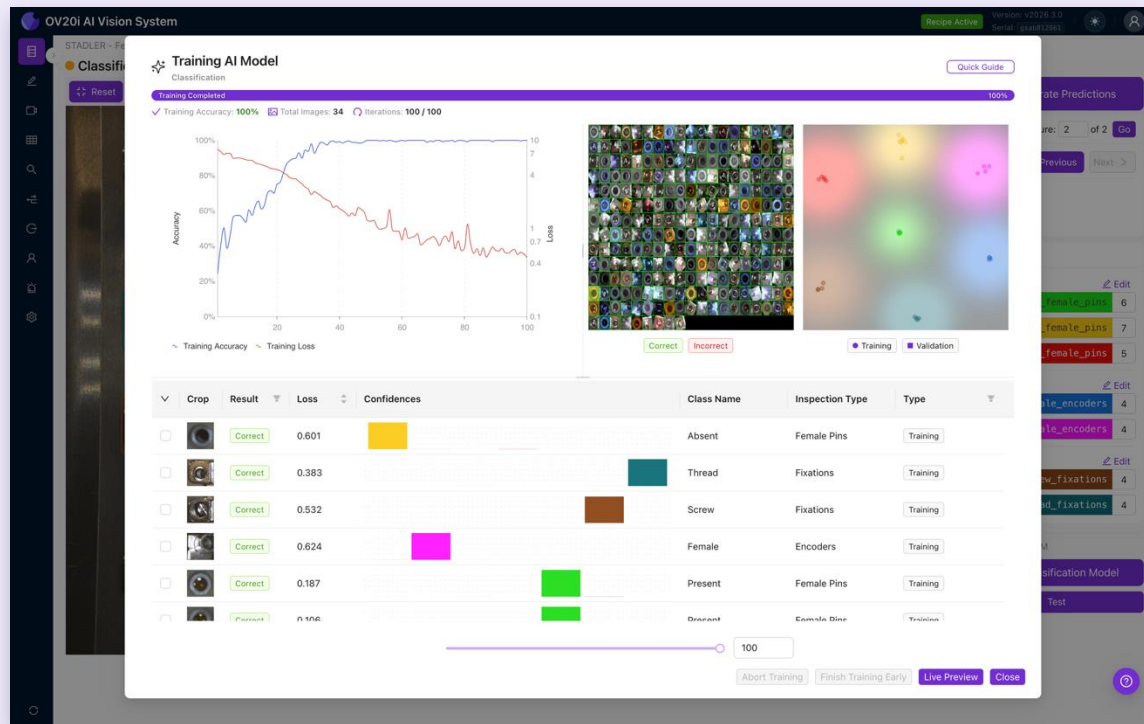
Training Dataset – Per-ROI View

Each ROI generates its own labeled crop that feeds the classification model. The OV20i shows every training crop with its class in one view — pins, encoders and fixations — making it easy to verify labels before training.



Step 5: Train the AI Model on the Camera

Training runs directly on the camera — no external PC, no software license. The model reached 100% training accuracy on the dataset built from one real image plus AI-generated defects, with per-crop confidences for every class.



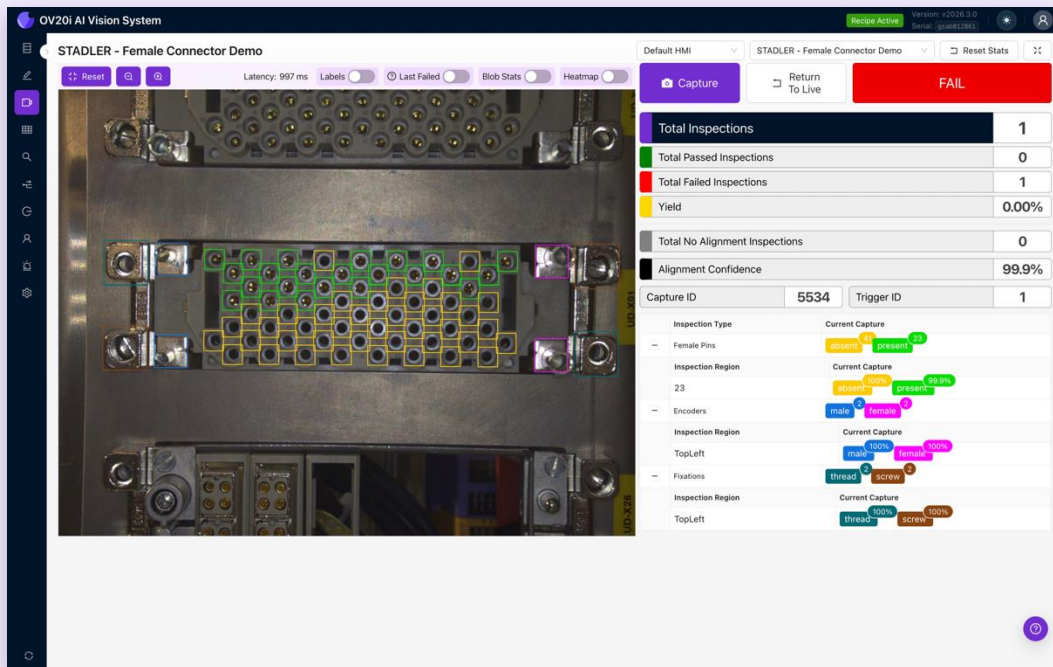
03

Custom Operator Dashboard

Built from scratch in 1 hour with the Node-RED AI flow creator

Out of the Box: Default Operator Dashboard (HMI)

Every recipe automatically gets a built-in HMI with live pass/fail, counters, yield and per-ROI results — zero configuration needed. For Stadler, we went one step further and built a fully custom operator dashboard.



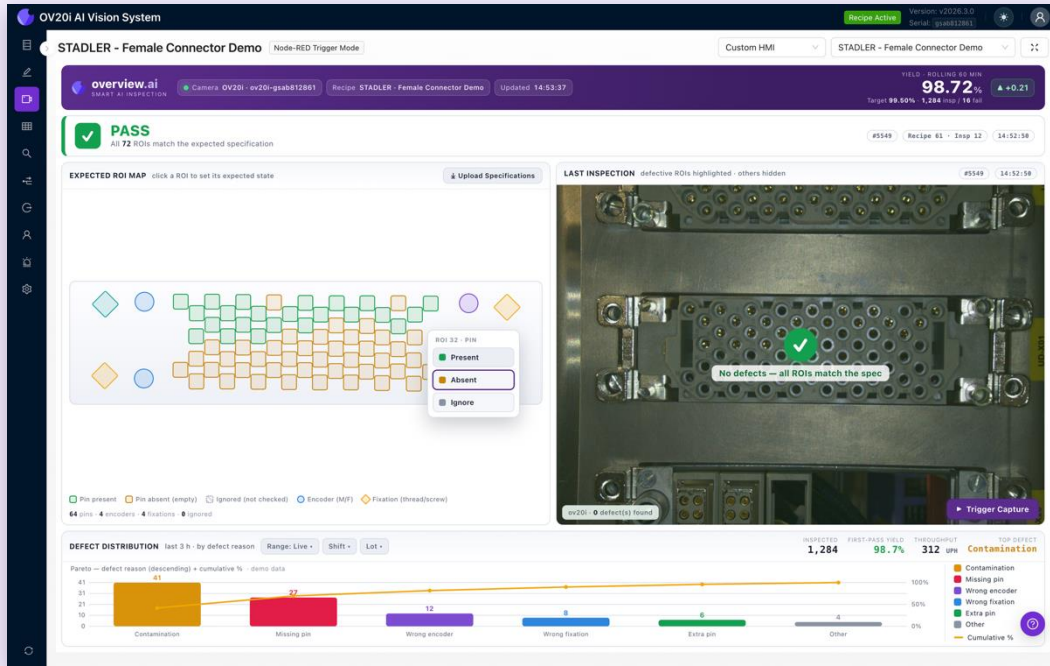
OV Auto-Integration Builder – Node-RED with AI Assistant

The OV20i embeds a full Node-RED environment with an AI Flow Assistant: describe the logic in plain language and it builds the flow. The complete Stadler dashboard — custom logic, specification handling and UI — was created from scratch in 1 hour. Included free with the camera.

The screenshot displays the OV Auto-Integration Builder interface, which is a Node-RED environment with an AI assistant. The interface is divided into several sections:

- Overview:** Includes tabs for 'Test Data Editor', 'Inspection & Verdict', 'Specification (CSV recipe)', and 'Dashboard'.
- Test Data Editor:** Contains a 'STADLER - Connector' section with a 'Trigger Simulated Capture' button. Below this are input fields for device_name, capture_id, recipe_id, inspection_id, inspection_time, image_url, and camera_id. There are also sections for 'alignment' (success, center_location_x, center_location_y, confidence, matched_angle) and 'roi' (rois, classification, segmentation).
- Inspection & Verdict:** This is the main workspace for building the flow. It shows a flow starting with 'All Block Outputs' connected to 'raw capture', which then feeds into '1 - Extract ROIs + meta', followed by '2 - Compare vs expected'. The flow also includes a 'Load static (deploy)' node and a 'Static: header + KPIs + Pareto' node.
- Flow Assistant:** A sidebar on the right with a 'Thinking' status and a list of steps: 'Reading your current flow', 'Querying recent defect data', 'Building Pareto chart', 'Adding chart to dashboard', and 'Deploying changes'. It also has a 'Modify "Inspection & Verdict"' section with 'Create', 'Chat', 'Flow', and 'Node' buttons.
- Dashboard:** At the bottom, there is a dashboard with a table showing inspection results. The table has columns for 'INSPECTION', 'I/O & LED', 'SAVE TO LIBRARY', 'METADATA', and 'DASHBOARD'. The data rows show 'NO RESULT' for all inspections.

Custom Dashboard – Operators Set the Connector Specs



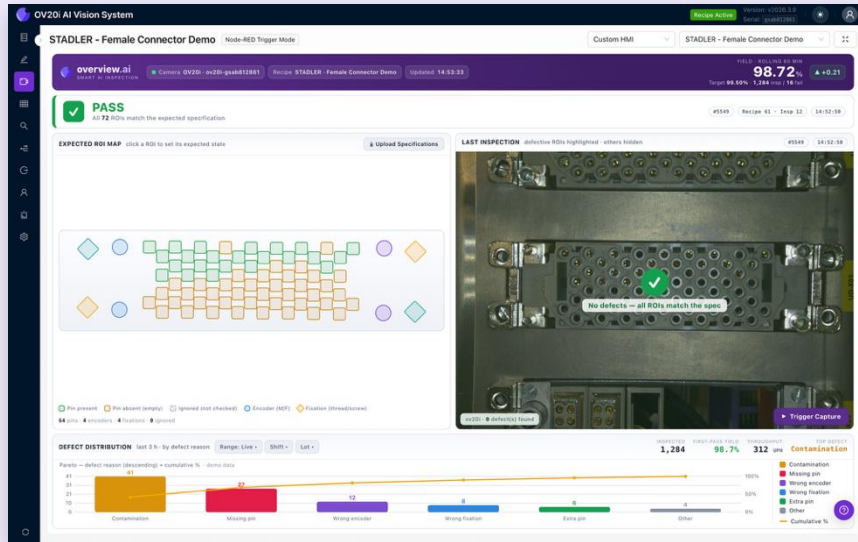
Why this matters for Stadler

- The operator selects the expected type of every female pin, encoder and fixation for the connector being assembled.
- Encoder and fixation type are the only safeguard against plugging a connector into the wrong adjacent position in the cabinet.
- Specifications can also be loaded from a CSV file or pushed automatically through the device API.
- No retraining needed — the same AI recipe is checked against the selected specification.

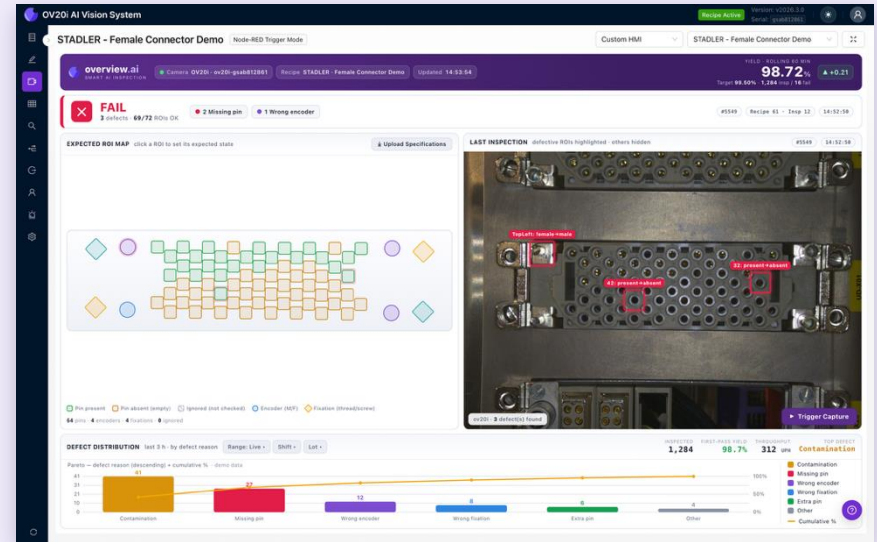
Live Inspection – Pass / Fail Against the Specs

Instant operator feedback

The dashboard compares every AI result with the selected specification and shows the verdict instantly, highlighting exactly which positions deviate from the expected configuration.



PASS – all positions match the specification



FAIL – mismatching positions highlighted

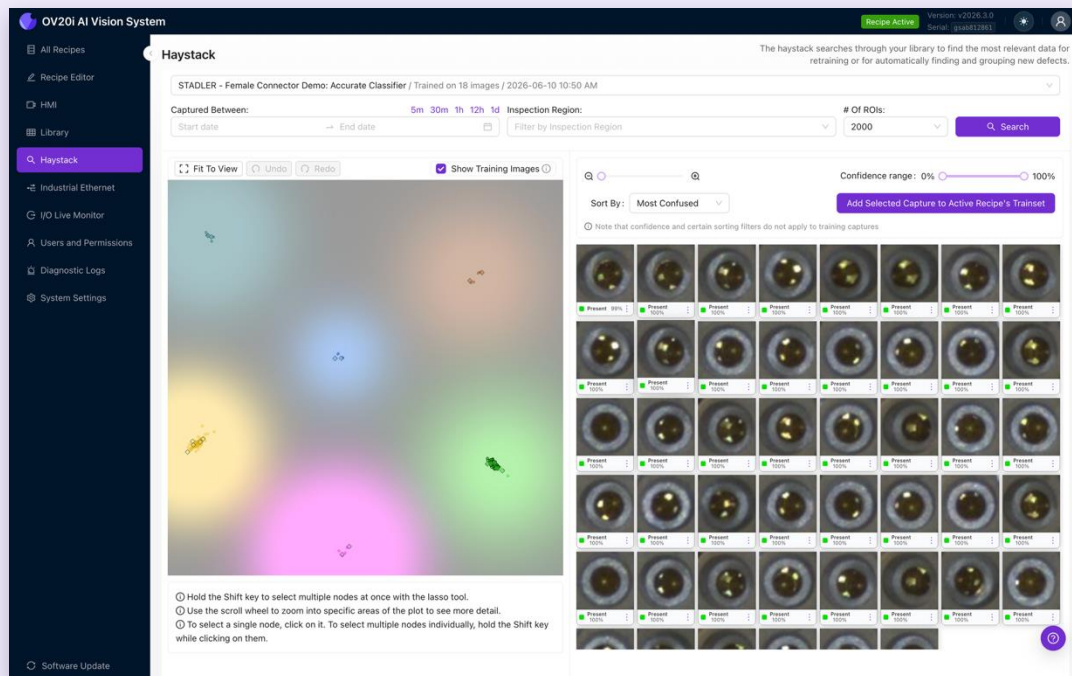
Image Library & Result History

All inspection captures are stored in the on-device library with full metadata, allowing engineers to review results, filter by recipe, and trace any inspection back to its source image (up to 300k images on device).

The screenshot displays the OV20i AI Vision System interface. On the left is a sidebar with navigation options: All Recipes, Recipe Editor, HMI, Library (selected), Haystack, Industrial Ethernet, I/O Live Monitor, Users and Permissions, Diagnostic Logs, and System Settings. The main area is titled 'Library' and shows a grid of inspection captures. The top of the library view includes filters for Capture #, Trigger ID, and Captured Between (Start/End date), along with buttons for 'Filter by Notes', 'Filter by Recipe', and 'Filter by Pass / Fail'. Below these are 'Advanced Filters: Metadata, Inspection Type, and Inspection Region' and a 'Search' bar. The grid shows several captures, with #5535 highlighted. To the right of the grid, a large image shows a detailed view of a connector with yellow and green bounding boxes. Below this image, the details for 'Capture #5535 from STADLER - Female Connector Demo' are shown, including a 'FAIL' status, trigger ID, capture time, process time, and download links. The 'Capture Notes' section is empty. Below the notes is a table with inspection results.

Inspection	Confidence	Class
Encoders		
2 Female 2 Male		
BottomLeft	100%	Male
BottomRight	100%	Female
TopLeft	100%	Male
TopRight	100%	Female
Female Pins		
41 Absent 23 Present		
2	100%	Present
3	100%	Present
4	100%	Absent
5	100%	Present

Haystack – Automatic Model Improvement

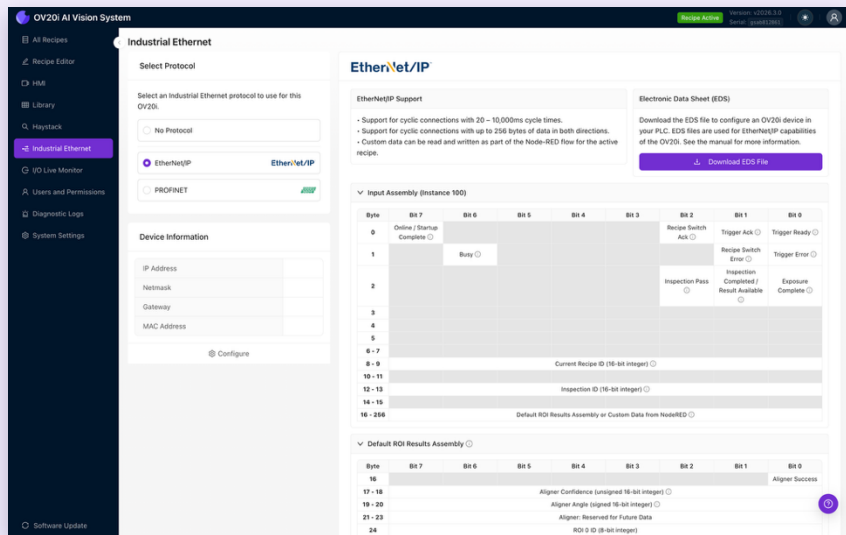


Find never-before-seen defects

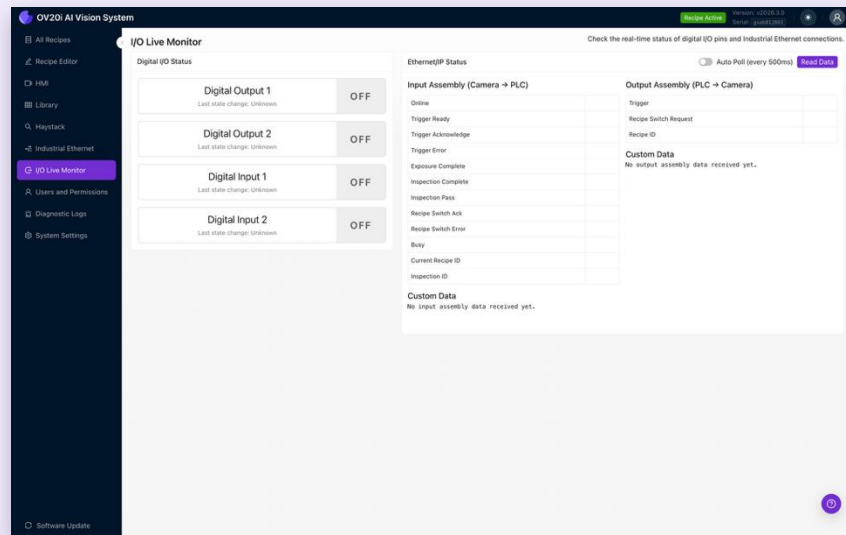
Haystack scans the on-device library to highlight images that don't match the trained distribution — typically new defect types or accuracy-improvement opportunities.

Selected outliers can be added back into the recipe's training set in one click — the model keeps getting better in production.

Industrial Connectivity & Integration



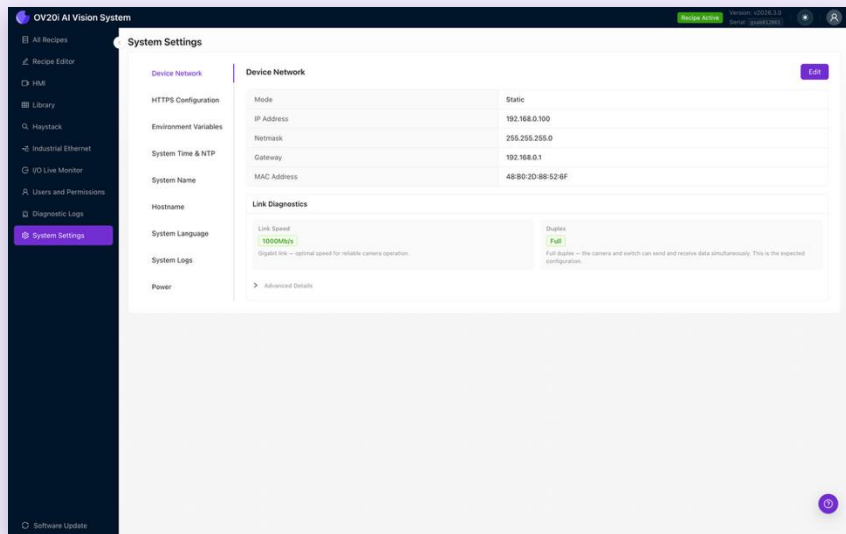
EtherNet/IP Protocol



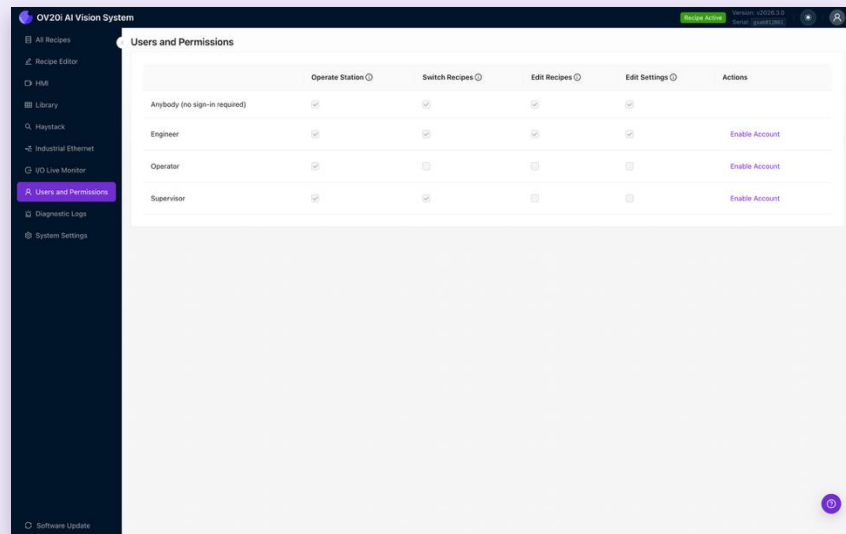
I/O Live Monitor

The OV20i supports EtherNet/IP, PROFINET, Modbus, MQTT, FTP, OPC-UA and most industrial protocols. 2× GPIO inputs, 2× outputs (5V to 24V) for direct PLC integration.

Built for Easy Commissioning



Network Settings & Link Diagnostics



Users & Permissions

Static IP configuration, link diagnostics and on-camera network debugging make commissioning fast. Role-based permissions (Operator, Engineer, Supervisor) control who can operate the station, switch recipes or edit settings.

Thank You

We look forward to helping Stadler St. Margrethen implement AI vision inspection on their production lines.

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